

RAG Platform – System Architecture v1

Tech stack:

Ollama llama3.1 LLM + Ollama nomic-embed-text + Chroma
FastAPI + PaddleOCR PP-OCRv6 + pdfplumber + pypdfi

5-Layer stack: Client / Edge / API Gateway / Service Layer / Storage + Cross-cutting Observability

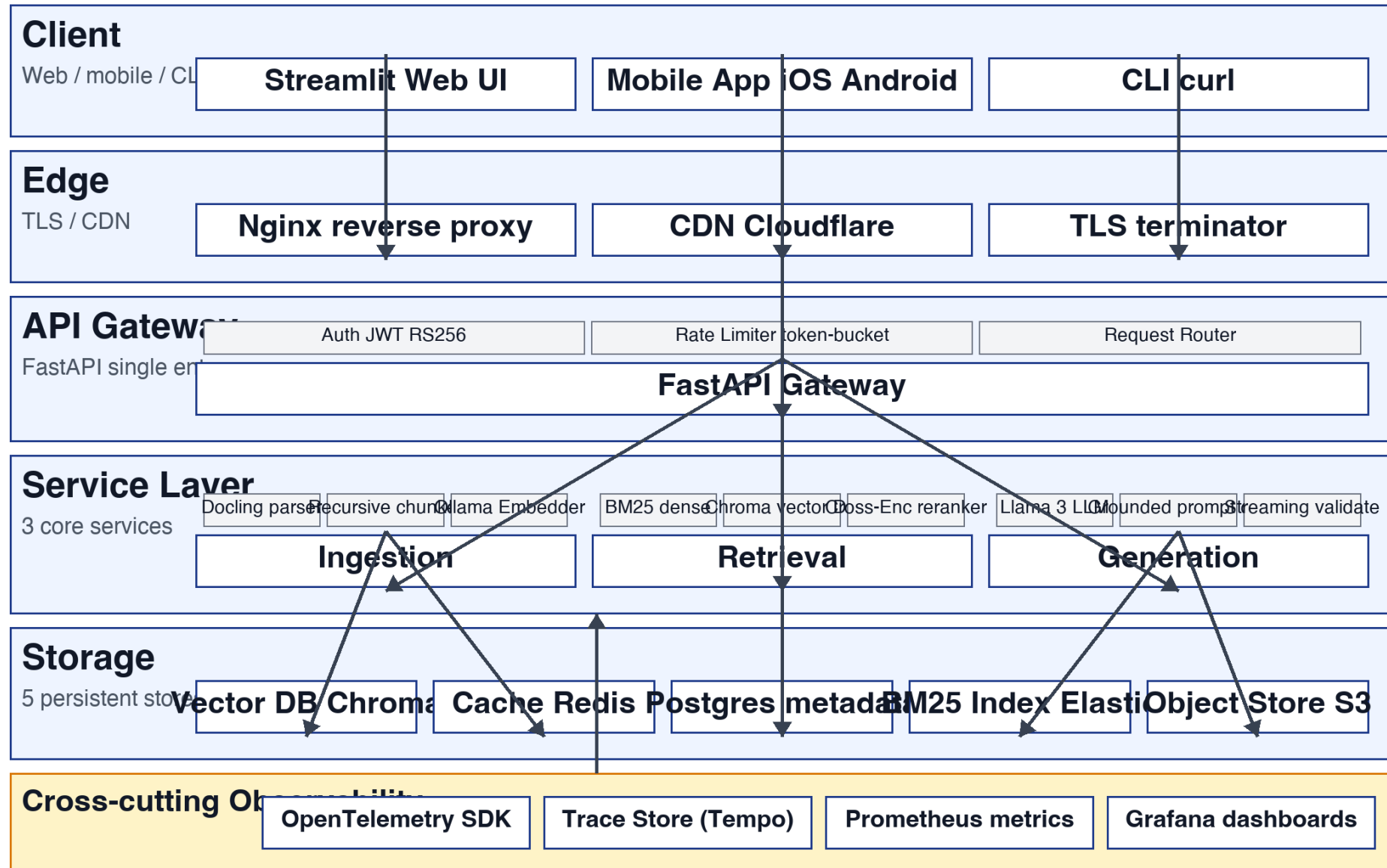


Figure 1: 5-Layer RAG Platform Reference Architecture – 2026

Component Responsibility Matrix

Per-layer Component / Responsibility / Tech Stack — v1 reference.

Layer	Component	Responsibility	Tech Stack
Client	Web UI	User chat + sources panel + diagnostics	Streamlit 1.x + httpx
Client	Mobile App	On-the-go Q&A via push notifications	React Native + AsyncStorage
Edge	CDN + TLS	Static caching, TLS termination	Cloudflare, Let's Encrypt
Edge	Nginx	Reverse proxy + edge rate-limit	Nginx 1.25 + lua-resty
API Gateway	FastAPI Gateway	Single entry point, OpenAPI spec	FastAPI 0.115 + Pydantic v2
API Gateway	Auth (JWT)	Issue + verify RS256 tokens	PyJWT + JWKS rotation
API Gateway	Rate Limiter	Token bucket per IP + per user	slowapi + Redis backend
Service / Ingestion	Multi-format Loader	PDF/HTML/DOCX/XLSX + OCR fallback	pdfplumber, BeautifulSoup, paddleocr
Service / Ingestion	Chunker	Recursive + Markdown-heading split	kbchat chunking 0.4
Service / Ingestion	Embedder	Vectorize chunks to 768-dim	Ollama nomic-embed-text
Service / Retrieval	Hybrid Retriever	BM25 + dense + RRF fusion	rank_bm25 + Chroma + RRF k=60
Service / Retrieval	Reranker	Cross-encoder score refinement	BGE-reranker-base (local)
Service / Generation	LLM	Grounded generation + citation builder	Ollama llama3.1 8B
Storage	Vector DB	Persist vectors + cosine search	Chroma PersistentClient
Storage	Cache	Hot Q/A cache + rate-limit counters	Redis 7 (sorted sets)
Storage	Postgres	Documents, users, ACL, audit log	Postgres 16 + pgvector mirror
Storage	Object Store	Raw uploads + derived artifacts	S3 / MinIO
Cross-cutting	OpenTelemetry	Trace every hop, export OTLP	opentelemetry-python + OTLP
Cross-cutting	Grafana	Latency / error / cost dashboards	Grafana 11 + Prometheus

Table 1: Per-component responsibilities and tech stack.

Data Flow Narrative

End-to-end pipeline: Ingestion, Query, Generation — with latency budget.

Ingestion Pipeline (offline, batch or watch-folder)

1. Watch folder or S3 bucket receives a new PDF, HTML, DOCX, or XLSX upload.
2. Loader parses each file. PDFs go through pdfplumber for text, tables, and image bboxes.
3. Each image region is rendered to PNG via pypdfium2 (scale=2.5), cropped by bbox, then OCR'd with PaddleOCR PP-OCRv6.
4. Chunking: recursive character splitter with overlap 200, or markdown-heading split for structured docs.
5. Embedding: each chunk is vectorized to 768-dim via Ollama nomic-embed-text, batched 32 at a time.
6. Persistence: vectors land in Chroma PersistentClient; metadata in Postgres; raw uploads in S3.
7. OpenTelemetry span closes; trace_id recorded in audit log for post-hoc RAGAS evaluation.

Query Pipeline (online, user-initiated)

1. Web UI sends the user question with JWT to FastAPI Gateway.
2. Auth middleware verifies RS256 token against JWKS; Rate Limiter checks token bucket (Redis).
3. Retrieval runs Hybrid Retriever: BM25 (rank_bm25) and Dense (Chroma) in parallel, fused via RRF k=60.
4. Top-20 candidates fed to Cross-Encoder reranker (BGE-reranker-base local), re-ranked to top-5.
5. Context Packer assembles passages with token budget 4000, de-duplicates near-duplicates.
6. Retrieval result + question forwarded to Generation service.

Generation Pipeline (online, streaming)

1. Grounded prompt template includes system instructions, retrieved passages, and citation slots.
2. LLM (Ollama llama3.1 8B) streams tokens; citation builder attaches [n] markers to source passages.
3. Response validator rejects answers lacking citation or containing prompt-injection markers.
4. Final answer + sources returned to Web UI; metrics emitted to OpenTelemetry.
5. Audit row written to Postgres with trace_id for end-to-end traceability.

Latency Budget (target p95)

Auth + Rate Limit 5ms | Hybrid retrieval 120ms | Rerank 80ms | Context pack 20ms | LLM first token 350ms | Total p95 less than 700ms.